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Will and Awareness

Free will is a concept deeply examined within both philosophical and psychological domains, though each approaches it with distinct lenses. Philosophically, free will concerns the ability of individuals to make choices freely, with various schools of thought offering different interpretations. From a psychological perspective, free will is analyzed in terms of cognitive processes and behaviors. Researchers delve into how choices are made, distinguishing between conscious deliberation and subconscious influences. Neuroscientific advances, such as those coming from brain activity studies, question the sequence and implication of conscious will in decision making processes. Moreover, the belief in free will also impacts our moral judgments and shapes societal attitudes toward issues (war, crime, empathy, etc).

Scientifically, the phenomenon knows many names. It can be addressed as cognitive control, strategic control, voluntary control, controlled processes, etc. Psychological investigations into cognitive control focus on conscious deliberation and the extent to which it plays a role in decision making versus subconscious or automatic processes, highlighted in Jacoby's influential paradigm. Neuroscience has contributed to the debate through studies of the brain, such as Libet's experiments, which call into question the timing of conscious decisions relative to brain activity. Findings in these areas have fueled discussions about whether the feeling of voluntariness is due to rationalizations after the brain has already initiated action. Psychological research additionally seeks to look into the dynamics of self control as a component of free will, such as the ability to delay gratification, resist temptations, and override impulsive behaviors.

The focal question of this paper is if free will, otherwise known as cognitive control, is exclusively conscious in the human mind? If this is not the case, to what extent is this exclusivity partial to the human mind? In exploring aforementioned psychological work by Jacoby and Libet, as well as good sensed rebuttal from psychologist van Gaal, this paper seeks to unearth and analyze evidence on the inclusivity or exclusivity behind free will in the conscious mind. Can access consciousness operate on conscious inputs or unconscious inputs?

Larry L. Jacoby is a renowned cognitive psychologist whose research interests lie in the study of human memory, specifically the mechanisms underlying different types of memory processes. Jacoby's work has been instrumental in distinguishing between explicit, declarative memory and implicit, non-declarative memory. Explicit memory involves conscious, intentional recollection of previous experiences—like recalling a fact or an event. Implicit memory, in contrast, is revealed by a change in behavior or performance that does not require conscious recollection—like performing a task more efficiently due to prior exposure, without the conscious awareness of such previous exposure.

The fundamental hypothesis underlying the Process Dissociation Procedure (PDP), the focus of Jacoby's 1991 publication, is that cognitive tasks can be influenced by both controlled (conscious, effortful) and automatic (unconscious, involuntary) processes, which can operate independently of one another. The logic of PDP is based on the assumption that these processes can be mathematically dissociated based on their different contributions under specific experimental conditions.

Jacoby designed PDP to provide a quantitative method for separating these two kinds of memory contributions when analyzing experimental data. Before the development of PDP, researchers found it challenging to differentiate between consciously controlled memory processes and automatic memory influences in tasks. This distinction is crucial because it helps researchers understand the various underlying mechanisms governing memory performance and how different conditions, disorders, or interventions might affect each type of memory differently. The PDP process typically involves presenting tasks under inclusion and exclusion conditions. In the inclusion condition, participants are asked to complete a task (ex. word completion) using all available information, which leverages both implicit and explicit memory. In the exclusion condition, participants are instructed to complete the task without using certain specified information, thereby intending to suppress explicitly remembered items. The differing performance across these conditions can be used mathematically to dissociate the contribution of controlled and automatic processes.

By developing this procedure, Jacoby was able to empirically test hypotheses about implicit and explicit memory processes and provide a framework for understanding the interplay and independence of these processes. This has had profound implications for studying not just memory in healthy individuals, but also changes in memory due to aging, neurological diseases, and psychological disorders. PDP aims to isolate and quantify the contributions of controlled and automatic processes. For example, if participants are able to exclude previously learned information as instructed in the exclusion condition, it suggests that controlled processes are overriding the automatic ones. The scores from these conditions can be used in a mathematical model to estimate the strength of the controlled and automatic processes. The resulting estimates reflect the degree to which each type of process influences task performance. Hence, PDP provides a way of quantifying and understanding the interplay between different types of memory processes that might otherwise be conflated in cognitive tasks. This is the means by which PDP partitions overall performance into conscious and unconscious components: by instructing participants differently in the inclusion and exclusion tasks and then applying a simple mathematical model to their responses.

In the context of subjective threshold paradigms, PDP features in discussions about subjective unconsciousness. When applied in subjective threshold paradigms, PDP is used to estimate the contributions of unconscious processes to overall performance. After accounting for performance that can be attributed to controlled, conscious processes, the PDP assumes that any remaining performance above chance level reflects unconscious processing. In this sense, the "subjectively unconscious" component refers to the proportion of cognitive processing that impacts behavior without entering conscious awareness, as inferred indirectly through performance on cognitive tasks. In terms of implications for free will and awareness, using PDP allows researchers to indirectly measure the extent of unconscious influences on decision-making, suggesting that a significant amount of our cognitive activity, including that which informs "free" choices, may be operating outside of conscious awareness. This understanding emphasizes that while we may feel our

decisions are freely made, they can be significantly shaped by unconscious factors, challenging the traditional view of free will as a wholly conscious deliberation. Moreover, these measurements emphasize that the subjective experience of being unaware doesn't imply the absence of cognitive processing; rather, it provides a framework for understanding how unconscious processes can still guide or influence behavior. This has important consequences for how we think about autonomy, responsibility, and the nature of our conscious experience.

The unconscious component estimated by PDP is argued to be unconscious based on the presupposition that certain cognitive processes can operate without reaching the level of conscious awareness, yet still influence behavior. This framework stems from the dual-process theory that posits the existence of controlled and automatic cognitive process. Reasons included the automatic processes are believed to be involuntary, fast, and operate without the need for conscious intervention. They can be triggered by certain stimuli or tasks and take place without the person's conscious awareness. Additionally, PDP aims to dissociate the contribution of conscious and unconscious processes by creating conditions wherein the participant's performance is attributable to one of these processes over the other. For instance, when participants are asked to complete a task that they can do without having to actively remember or be aware of the information (e.g., completing a word stem with a word that was previously studied), any contribution to performance above chance level that cannot be attributed to conscious recall is inferred to be an automatic, unconscious process. Further, by comparing performance on inclusion and exclusion tasks, PDP estimates the strength of the controlled processes. The automatic, or unconscious, component is deduced from the performance that exceeds chance level once the conscious, controlled processes have been accounted for. Also, participants often report no conscious awareness of the stimuli influencing their behavior in a task, which provides subjective evidence that supports the classification of a process as unconscious. When participants can complete a task without being able to report the use of explicitly remembered information, it suggests that unconscious processes are at play. Finally, A hallmark of conscious processing is intentionality and deliberate choice. Unconscious processes, in contrast, are marked by a lack of intention. If a participant's action appears to be guided by something other than a deliberate decision—such as an implicitly learned response pattern—it is considered an unconscious influence.

Thus, the unconscious component in the PDP framework is argued to be unconscious because it operates without conscious awareness, intention, or control, as per the evidence that emerges from the differential performance in the task's conditions and subjective reports. It encompasses the processing that influences behavior invisibly, beyond the scope of introspection and voluntary control. Despite these critiques, the introduction of the PDP represented a significant advance in cognitive psychology and provided a valuable tool for researchers. It helped to stimulate ongoing debate about the nature of memory processes and led to further refinements of the method and the development of new techniques for investigating the complex interplay between conscious and unconscious cognition.

Another psychologist, Benjamin Libet, a pioneering researcher in the field of neurophysiology is best known for his experiments in the 1980s that explored the timing of conscious decision making, particularly in relation to the neural basis of free will. Libet's most famous experiment involved the decision to move method, also known as the Libet experiment, which sought to examine the temporal relationship between the conscious intention to perform a voluntary action and the brain's initiation of that action. Libet's interest in decision-making was primarily focused on the timing and neurological basis of conscious volition and the initiation of voluntary actions. His fascination lay in exploring the relationship between brain activity that precedes a conscious decision and the subjective experience of making a decision. Libet was particularly interested in the implications his findings would have for understanding the nature of free will.

The motivation behind designing the decision to move experiment was to empirically investigate the premise that conscious intentions directly lead to voluntary actions—a hallmark claim in discussions of free will, especially in the space of philosophical debate. Traditional views on free will suggest that a conscious decision precedes and causes our voluntary actions to occur. If this were the case, Libet expected that the conscious intention to act would emerge before

any related neural activity in the brain. However, prior neuroscientific research had identified brain activity, known as the readiness potential (RP), which appeared to precede voluntary movements. Libet set out to design an experiment that could capture the precise timing of this brain activity and relate it to the subjective report of conscious intention. By having participants perform a simple, spontaneous voluntary movement while observing a clock to report the moment of their conscious intention to move, Libet aimed to establish a timeline of events leading up to a voluntary action. Simultaneously, Libet measured the participants' brain activity through an electroencephalogram (EEG) to detect when the RP occurred.

The RP began significantly before participants reported the first awareness of their intention or urge to move. On average, the RP started approximately half a second before the action, and about a few hundred milliseconds before participants became conscious of their intention to act. The groundbreaking aspect of this experiment was that it provided empirical data connecting the subjective experience of decision-making with measurable brain activity. To Libet's surprise, his findings suggested that the RP preceded the conscious intention to move by several hundred milliseconds. Libet's interpretation of these results led to his hypothesis that unconscious processes in the brain might set the stage for a voluntary action well before we become aware of our intention to act. Yet, he also proposed that there is a brief window in which the conscious self could "veto" the action after the onset of the RP but before the movement is executed. He called this the "veto possibility," which, according to Libet, preserved a role for conscious free will, though in a more limited capacity than traditionally conceived.

The decision to move experiment and its implications continue to be a subject of much debate. Researchers have challenged the experiment's methodology and interpretations of the results, and there have been further studies aimed at expanding upon or refuting Libet's conclusions. In 2007, Banks and Pockett raised several objections to Libet's findings and interpretations, reflecting broader concerns within the scientific community. One major objection concerns the method by which participants report the time of their conscious intention (“W” moment). Banks and Pockett highlighted potential inaccuracies in using the clock to report this moment. They argued that the subjective process of noting the position of a dot on a fast-moving oscilloscope to time a conscious experience could introduce significant error, leading to unreliable data about the timing of awareness. Another objection pertains to the assumption that the RP recorded by EEG is directly related to the initiation of the specific, voluntary action being studied. Critics like Banks and Pockett noted that RP could represent general preparatory brain activity rather than activity specific to the actual decision to move. Therefore, the RP might not be the neural marker of intending to move at a precise moment. Libet's interpretation that while the unconscious brain activity might initiate an action, conscious will could still veto it was also contested. Banks and Pockett argued that this "veto" does not necessarily constitute evidence for free will because the veto itself might be initiated unconsciously, with the consciousness only becoming aware of the action post hoc. An objection also arises from the possibility that the subjective experience of time might be distorted in the experimental setup. Participants might perceive and report the intention before the actual action is taken, not because consciousness initiates the action, but because the subjective experience of time is compressed or elongated under the conditions of the experiment. It was suggested that the RP might sometimes be the result of neural noise or non-specific activity not directly related to the action, which challenges the concept that the RP directly indexes the decision to move. Banks and Pockett, along with other critics, have questioned whether the findings from Libet's laboratory tasks can be generalized to more complex, real-world decisions. The simplistic nature of deciding to flick a wrist might not be representative of the multifaceted decision-making processes that occur in everyday life situations.

These objections reflect deeper debates around the methodologies, interpretations, and implications of Libet's work, which have had a considerable impact on our understanding of conscious will and the nature of voluntary action. In subsequent years, researchers have continued to explore these objections, leading to refinements of experimental

techniques and theoretical models. The dialogue between experimental findings and philosophical interpretations continues to evolve, demonstrating the complexity and depth of inquiry into the workings of the human mind.

Is there evidence for unconscious free will or cognitive control? In 2012, Sander Van Gaal and colleagues conducted a study investigating the neural mechanisms behind error detection, task switching, and response inhibition to dive into such curiosities. This study was designed to explore how the brain processes and corrects spontaneous errors, even when those errors do not reach the level of conscious awareness. A particular focus on Van Gaal's approach to response inhibition should be highlighted as the no-go tasks reframed how many viewed the issue.

No-go tasks are a form of psychological experiment used to study response inhibition, which is the ability to suppress actions that are inappropriate or no longer required. These tasks are designed to measure an individual's ability to withhold a motor response in the face of stimuli that usually trigger that response. No-go tasks have a simple structure where participants are typically presented with stimuli and are asked to perform a specific action (like pressing a button) when they see certain "go" signals. However, they must inhibit this action when a different "no-go" signal appears. Go trials are frequent and require a habitual response from the participant. No trials are less frequent and require the participant to withhold the response they have been trained to make during the go trials. To measure inhibition the crux of the task is the participant's ability to inhibit the prepotent (or habitual) motor response during the no-go trials. The response times, as well as the accuracy of the participant's performance, are recorded to assess their inhibitory control. They were required to follow certain rules that sometimes changed, thereby integrating elements of task switching and response inhibition. The study distinguished between errors that participants were consciously aware of and those they did not consciously notice. This differentiation was critical, as it aimed to investigate the cognitive and neural processes underlying error detection. Using functional magnetic resonance imaging (fMRI), Van Gaal and colleagues observed brain activity associated with both aware and unaware errors. This imaging allowed researchers to see which brain regions were activated during error processing. From a neuroscientific perspective, no-go tasks are associated with the activation of specific brain regions, particularly those involved in the executive control of actions, such as the prefrontal cortex. The neural correlate often measured in such tasks is the No-Go P3, an event-related potential (ERP) component observed in EEG recordings. The results showed that some brain regions involved in error processing, prefrontal cortex P3, were activated regardless of whether participants were aware of their mistakes. However, certain areas, especially in the medial frontal cortex, showed amplified activity only when participants consciously recognized their errors. The study suggested that the brain can register and process errors at an unconscious level, potentially leading to an automatic error correction system. Nonetheless, for more complete error processing and potentially to adaptively adjust future behavior, conscious awareness might play a critical role.

This research contributed to the broader understanding of cognitive control processes and the neural basis of human error monitoring. The effectiveness of our inhibition processes, as tested by no-go tasks, could be seen as a measure of one's capacity to exercise free will. If participants in no-go tasks can successfully inhibit a response to a no-go stimulus, it suggests the presence of an awareness of the task demands and a voluntary control to override an automatic response. It provided evidence that the brain has the capacity to detect and process errors without conscious awareness but also highlighted the additional mechanisms that come into play when errors are consciously perceived. The study, therefore, supports the notion that both conscious and unconscious processes are involved in the complex task of cognitive control and behavioral adjustment. Van Gaal highlights that there may be very important limits to his study. Perhaps the time frame takes an effect in controlling long lasting or future responses. Perhaps spontaneity takes an effect in the initiation of cognitive control. Other authors, Snodgrass, ponder if unconscious control can only be possible when the task is consciously automatized.

The works above all point to the notion that awareness involves a dual relationship between consciousness and unconsciousness. That there are quantifiable means to differentiate them as they show mixed presence in participant

results. That unconscious processes in the brain set the stage for a cognitively controlled action well before we become aware of our intention to act. That the brain has the capacity to detect and process errors without conscious awareness. Ultimately, in my perspective, the question of free will being exclusively consciousness is a repacking of the question regarding the nature, range, and limitations of access consciousness. As the reflective branch of consciousness, it is inherently tied to free will as a notion and heavily inspired by individualism and personal preference. It also draws upon the notion of subjectivity and objectivity in regard to consciousness research. In review of both these topics of consciousness– Phenomenal vs Access and Subjective vs Objective – the consensus was awareness presents itself as a blend of both conscious and unconscious hierarchies all subject to higher level systems of thought. I once made the argument that no brain process can be entirely phenomenal or entirely access as even if a being has a fraction of a fraction of presence of access consciousness and is near fully existing within the physical and phenomenological realm, their fractional access–consciousness exist all the same and if ever expanded on will remember its minimization. Vise versa, as if a being is living fully in the access–conscious, their physical reality is minimized, however there are less dire consequences to the lack of physical experience once a thinker has true perspective. This same notion is expressed in the objective methods of exploring consciousness and the concept of 'above zero' awareness. The approach arguing that typical above–chance performance under subjective conditions could plausibly reflect weak results but nonetheless conscious perception. To rule out this possibility of perceptual sensitivity of zero, there's a requirement for far more stringent stimulus conditions. Then if one still gets evidence of cognitive processing, one can conclude such effects are truly a reflection of zero stimuli unconsciousness.

Here, I introduce the notion that free will, like access consciousness, is first exclusive before inclusive. This is to say a human's subconscious can exclusively decide to carry inclusive processes. Willfully decide to carry ignorance about the ways certain processes function in the body. My assumption is that this is likely to save energy, as it is improbable to ruminate on every singular system your brain regulates while trying to engage in a social capital and be present in one's livelihood. It is similar to the physical concept of creating a routine. Maybe it takes some intentionality to fall asleep and wake at the same time every day, eat on a even cycle, work out, and have a balanced lifestyle, but with disciple, eventually these things that may once have seem tedious and overwhelming can become a humans natural routine they must follow to feel well and at homeostasis themselves. It became their automatic routine that does not take much forethought, however developing such a routine and remaining consistent with it, did. This is how I view the exclusivity and inclusivity problem in regard to free will. Human brain processes have been around for generations and it is only natural to assume that the organ developed unconscious and heavily ancestral pathways based on its previous experience.

The collective insights from process dissociation procedures (PDP), Benjamin Libet's "decision to move" experiments, and studies on response inhibition like no–go tasks offer a multifaceted view of will and awareness. Research in these areas has informed our understanding of the relationships between cognitive processes, neural activity, and the subjective experience of control and decision–making. These studies underscore that decision–making is highly complex, involving both conscious and unconscious processes. Our awareness of some decisions and responses can lag behind the initiation of neural processes that lead to action. PDP and no–go task studies illustrate the delicate balance between automatic (implicit) processes and controlled (explicit) processes. This interplay is a critical factor when considering actions we describe as free or voluntary. Libet's experiments raise questions about the timing of conscious intentions relative to neural events. This idea supports a model of free will where consciousness has the power to stop or modify actions but may not initiate them.

In conclusion, the findings from these lines of research do not entirely resolve the debates regarding free will and conscious awareness. Instead, they enrich the conversation with empirical data that both challenge and enlighten traditional philosophical perspectives. Free will and the role of consciousness in voluntary action appear more nuanced than previously believed, opening up new avenues for understanding the cognitive neuroscience of human choice.

Works Cited

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